Snap-Conditioned Retrieval for Legal RAG: A Four-Benchmark Bottleneck Study

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Abstract

Legal retrieval-augmented generation is often reported as a single accuracy number: retrieve documents, generate an answer, and compare against a no-RAG baseline. Our experiments show that this is the wrong abstraction for legal QA. We study snap-conditioned HyDE methods across four legal benchmarks: BarExamQA, HousingQA, LegalBench-SCALR, and CaseHOLD. Snap-conditioned HyDE first produces a tentative legal analysis, uses it to generate a hypothetical retrieval passage, and then answers with retrieved evidence. On full-corpus BarExamQA, `rag_snap_hyde` improves Gemma 4 26B from 78.08% to 81.17% (+3.09pp, McNemar $p = 0.0130$) and Gemma 4 E4B from 58.49% to 62.18% (+3.68pp, $p = 0.0129$). However, the broader legal results are mixed: BarExamQA is retrieval-depth flat, LegalBench-SCALR is candidate-depth sensitive then saturates at top-5, HousingQA has a directional top-10 statutory-retrieval signal, and CaseHOLD improves gold-option retrieval without a reliable answer lift. We therefore frame snap-HyDE as a useful method family and probe, not as a universal legal RAG solution. The main contribution of this project is a bottleneck-typed legal RAG analysis: retrieval depth, candidate-set size, answer anchoring, metadata filtering, and evidence-to-answer conversion must be separated before tuning methods.

1 Introduction

Legal QA is a natural target for retrieval augmentation. Many questions depend on statutes, holdings, jurisdiction, procedural posture, and precise rule exceptions that may not be reliably stored in a model's parameters. However, legal RAG is also a setting where more retrieved text can easily hurt. A passage can be topically related but legally incomplete; a holding can mention the same doctrine while being the wrong option; a state statute can be right in subject matter but wrong in jurisdiction. These errors are not just retrieval failures or generation failures. They are interactions between retrieval, reasoning, and answer selection.

Our initial project pursued an agentic legal RAG system: route the question to a corpus, decompose the issue, retrieve and summarize evidence per gap, judge sufficiency, and replan when needed. That design is realistic for legal research, but it is difficult to evaluate scientifically. If a full agent improves, the gain could come from better retrieval, more tokens, a second reasoning pass, or a lucky prompt. If it fails, the error could come from over-decomposition, bad gap detection, irrelevant evidence, or answer overwriting. We therefore moved from full-system complexity to atomic retrieval interventions.

The central method family is snap-conditioned HyDE. A "snap" is a first-pass answer or reasoning trace. A HyDE-style hypothetical passage is then generated from that snap and used for retrieval. The final answer sees the question, snap reasoning, and retrieved evidence. The intuition is legal: before asking the model to read external authority, ask it to state the issue and likely rule. Retrieval is then shaped by that legal frame rather than by the raw question alone.

The current legal-only evidence is not uniformly positive, and that is useful. BarExamQA shows a real full-corpus snap-HyDE lift, but top-1 versus top-5 retrieval is flat, so the lift should not be described as "more documents help." LegalBench-SCALR shows a strong top-1 collapse and top-5 saturation, which suggests candidate-set size matters more than pseudo-document query formulation. HousingQA has a promising top-10 signal, but a state-filter run failed due a metadata-casing bug and remains pending after repair. CaseHOLD now has repaired gold-option retrieval, and snap-HyDE raises gold retrieval from 16.0% to 47.0%, but answer accuracy only moves from 69.5% to 72.0% with $p = 0.4421$.

We exclude MuSiQue from the main report because it is not a legal benchmark. It remains a useful internal multi-hop mechanism check, but the final class report is framed around four legal benchmarks. This makes the narrative less numerically convenient, because the strongest two-call snap-HyDE lift in the repo is currently on MuSiQue. That is acceptable for the current stage. The goal is to identify which legal bottlenecks are real, then tune retrieval and HyDE quality toward those bottlenecks.

2 Related Work

Retrieval-augmented generation combines parametric generation with retrieved non-parametric evidence [Lewis et al., 2020]. HyDE generates a hypothetical document and embeds it for zero-shot retrieval [Gao et al., 2023], while Query2Doc expands queries with LLM-generated pseudo-documents [Wang et al., 2023]. FLARE retrieves during generation by using predicted future content as a query [Jiang et al., 2023]. Our method family inherits directly from this line: generated text is used as a retrieval handle, not as trusted evidence.

Adaptive RAG work asks when or how retrieval should be invoked. Self-RAG learns to retrieve and critique through reflection tokens [Asai et al., 2024]. Corrective RAG grades retrieved evidence and triggers correction when retrieval is unreliable [Yan et al., 2024]. Adaptive-RAG routes questions by complexity [Jeong et al., 2024]. Speculative RAG drafts multiple answers from retrieved subsets and verifies them with a larger model [Wang et al., 2025]. These systems motivate our future direction, but we do not claim to implement them here. Instead, we use their shared lesson: retrieval should be conditional on observed failure mode.

Legal NLP benchmarks increasingly show that legal RAG cannot be reduced to generic open-domain QA. LegalBench provides a broad suite of legal reasoning tasks, including SCALR holding selection [Guha et al., 2023]. CaseHOLD evaluates whether a model can select the holding referenced by a court-opinion context [Zheng et al., 2021]. Zheng et al. introduce BarExamQA and Housing Statute QA to evaluate realistic legal retrieval and answering workflows [Zheng et al., 2025]. LegalBench-RAG focuses on retrieving minimal relevant legal snippets [Pipitone and Alami, 2024], and Legal RAG Bench provides an end-to-end legal RAG benchmark with factorial retriever/generator decomposition [Butler and Butler, 2026]. RAGChecker similarly argues for fine-grained separation of retrieval and generation errors [Ru et al., 2024]. Our work follows this diagnostic turn: the same RAG method should be judged differently depending on whether the active failure is retrieval depth, candidate-set selection, metadata filtering, or evidence conversion.

We also considered hypothetical-representation work beyond HyDE. HyQE ranks contexts with hypothetical query embeddings [Zhou et al., 2024]. We did not implement a separate "HyRE" or hypothetical reasoning embedding index in this report. The implemented object is simpler: online snap-conditioned HyDE used as a legal retrieval probe.

3 Framework Design

3.1 Method ladder

The report uses an inheritance-style method ladder. Each method adds one capability where possible, so negative rows remain interpretable.

Method	What it isolates
`llm_only`	Parametric legal answer without retrieval.
`snap_only_in_final`	Whether a first-pass answer/reasoning trace helps without retrieval.
`rag_simple`	Standard retrieval plus answer generation.
`rag_hyde`	Question-conditioned hypothetical document retrieval.
`rag_snap_hyde`	Snap-conditioned HyDE retrieval plus final synthesis.
`rag_snap_hyde_1call`	Whether snap reasoning alone after ordinary retrieval is enough.
`rag_snap_hyde_2call`	Fixed-cost snap and HyDE passage in one first call, final synthesis in a second.
`multi_hyde_diverse`	Multiple generated retrieval passages for query diversity.
`subagent_rag`	Gap-routed report generation; currently used as a negative over-abstention control.

3.2 Snap-conditioned HyDE

Figure 1 shows the core method. The model first writes a snap answer or legal analysis. It then writes a hypothetical passage that states the likely controlling rule, fact, or holding. That passage is embedded and used for retrieval. The final model sees retrieved passages along with the original question and snap reasoning.

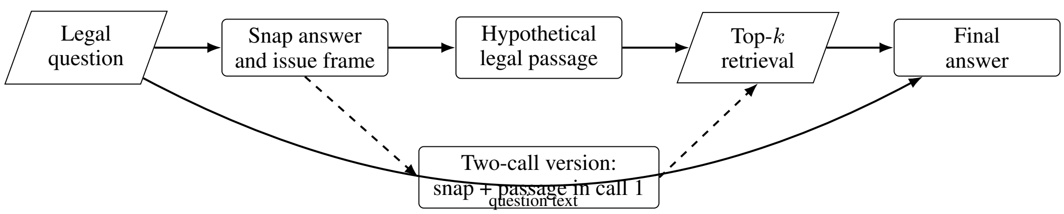

Figure 1: Snap-conditioned HyDE. The generated passage shapes retrieval; the retrieved corpus passages remain the external evidence used by final synthesis.

3.3 Bottleneck diagnostics

We use three families of diagnostics. First, top-k sensitivity tests whether answer accuracy depends on retrieving more than the single best passage. This is a retrieval-policy stress test, not pure context truncation, because the cross-encoder reranks a first-stage candidate pool before selecting the final top-k. Second, snap-only and HyDE-only ablations ask whether the gain comes from answer anchoring, generated-query retrieval, or their combination. Third, gold-retrieval and disagreement audits ask whether better retrieval actually converts into better final answers.

The four legal benchmarks map naturally onto different bottleneck hypotheses: BarExamQA tests answer anchoring under strong legal priors, HousingQA tests jurisdiction-specific statutory retrieval, LegalBench-SCALR tests small candidate-set holding selection, and CaseHOLD tests whether a retrieved holding can be converted into the correct answer option.

4 Experimental Setup

Benchmarks. BarExamQA contains 1,195 bar-exam multiple-choice questions with fact patterns and legal passages. HousingQA contains yes/no housing-law questions over state statutes. LegalBench-SCALR contains 571 Supreme Court holding-selection questions. CaseHOLD contains court-opinion contexts and five candidate holdings. We deliberately exclude non-legal MuSiQue from the main benchmark set, although it remains useful for internal multi-hop stress testing.

Models. The full-corpus BarExamQA rows use Gemma 4 26B-A4B and Gemma 4 E4B served through cluster logs. LegalBench-SCALR and CaseHOLD diagnostic rows use Llama 70B. HousingQA uses Gemma 4 26B. For paired claims, we compare runs on the same question slice and provider. The laptop Chroma checkout is useful for SCALR smoke tests, but it is not the availability boundary for this project: the cluster checkout contains the larger legal collections, so BarExamQA, HousingQA, CaseHOLD, and SCALR follow-up retrieval runs are launched through SLURM when they need populated corpora.

Metrics. We report accuracy for multiple-choice and yes/no datasets. Paired deltas use exact McNemar tests with b/c discordance counts. Retrieval diagnostics report gold retrieval only when the dataset's gold mapping and corpus collection are known to be valid. All headline rows in this draft were re-checked against detail logs in `reports/final_class_report/evidence_snapshot.md`.

5 Results and Analysis

5.1 BarExamQA: full-corpus snap-HyDE lift

BarExamQA is the strongest legal positive result. On Gemma 4 26B-A4B, `rag_snap_hyde` reaches 81.17%, compared with 78.08% for `rag_simple` (+3.09pp, b/c=124/87, $p = 0.0130$). On Gemma 4 E4B, the same method improves 58.49% to 62.18% (+3.68pp, b/c=172/128, $p = 0.0129$).

Table 1: BarExamQA full-corpus Gemma 4 26B-A4B method matrix, $N = 1195$.

Method	Accuracy	Δ vs `rag_simple`
`rag_simple`	78.08%	–
`golden_passage`	78.66%	+0.58pp
`rag_hyde`	78.91%	+0.83pp
`llm_only`	79.75%	+1.67pp
`snap_only_in_final`	80.59%	+2.51pp
`rag_snap_hyde`	**81.17%**	**+3.09pp**

The mechanism is not pure retrieval depth. On a paired $N = 200$ diagnostic, `rag_simple` top-5 is 82.5% and top-1 is 83.0% ($p = 1.0$). The full-corpus lift is better read as snap answer anchoring and evidence reconciliation. A legal MC model can have strong prior knowledge of the rule; retrieval then helps only if it is framed in a way that does not overwrite the right legal analysis with a partial snippet.

5.2 Four-benchmark legal bottleneck matrix

Table 2 is the compact result summary. It intentionally includes negative and directional rows. The current evidence does not show one method winning everywhere; it shows which legal bottleneck each dataset is probing.

The most important negative result is CaseHOLD. After repairing the holdings corpus and gold mapping, two-call snap-HyDE finds the gold holding much more often. If retrieval recall were sufficient, answer accuracy should improve sharply. It does not. This suggests an evidence-to-option bottleneck: retrieved holdings can still be paraphrased distractors, overly generic, or hard to map back to the exact candidate label.

SCALR gives a different holding-selection story. Top-1 is poor, top-5 fixes most of the loss, and top-10 adds no answer accuracy beyond top-5. This is not a generic "legal holdings ignore retrieval" result. It says SCALR needs a small candidate set, while CaseHOLD needs better evidence conversion.

HousingQA is the most important unfinished legal retrieval result. The top-10 lift is directionally positive, but the state-filter experiment failed because question states were display-case strings while Chroma metadata was lowercase. That is an infrastructure bug, not a method result. The fixed state-filter run is pending; until then, HousingQA supports only a cautious statutory-depth claim.

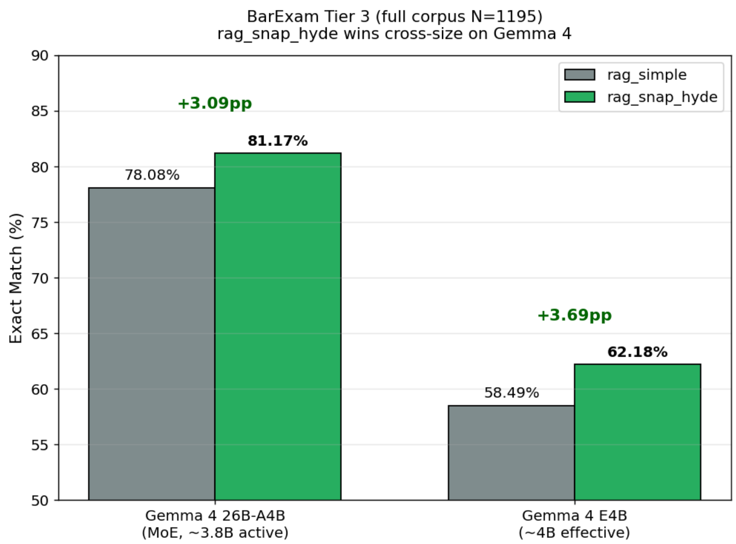

Figure 2: BarExamQA cross-size result. `rag_snap_hyde` improves over `rag_simple` on both Gemma 4 model sizes.

Table 2: Legal-only benchmark matrix. MuSiQue is excluded because it is not legal-domain data.

Benchmark	Snap-HyDE / method result	Depth or retrieval diagnostic	Current read
BarExamQA / Gemma 4	Full $N = 1195$: +3.09pp at 26B, +3.68pp at E4B	top-5 82.5% vs top-1 83.0%, flat	Legal MC is answer-anchoring dominated, not depth limited.
HousingQA / Gemma 4	`rag_snap_hyde_2call` 57.0% at k=5	top-1 50.5% to top-10 58.0%, +7.5pp, $p = 0.0722$	Directional statutory-depth signal; state filtering is pending.
LegalBench-SCALR / Llama 70B	`rag_snap_hyde_2call` 75.0% vs `rag_simple` 77.0%, NS	top-1 59.5%, top-5 77.0%, top-10 77.0%	Candidate-set limited until top-5, then saturated.
CaseHOLD / Llama 70B	Repaired pair: 69.5% to 72.0%, $p = 0.4421$	gold retrieval 16.0% to 47.0% under two-call	Better gold retrieval does not yet convert to reliable answer lift.

5.3 Gold passage is not an oracle in legal MC

The BarExamQA golden-passage control appears paradoxical: providing the gold passage reaches only 78.66%, below `llm_only` at 79.75% and `snap_only_in_final` at 80.59%. The reason is legal evidence granularity. A labeled passage can be topically correct but not decisive for the fact pattern; it can also highlight a distractor doctrine. Snap-first reasoning gives the model an initial legal frame before evidence reconciliation, which partly protects against over-reading one snippet.

5.4 Current follow-up runs

The next legal-only runs are now cluster-backed rather than laptop-limited. The active set is:

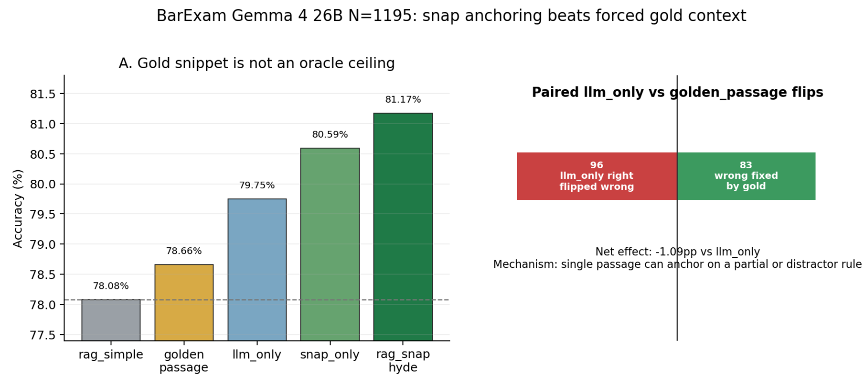

Figure 3: BarExamQA mechanism audit. A single gold passage is a noisy control rather than an oracle ceiling; snap-first reasoning can recover some cases where forced context distracts the model.

Job	Purpose
SLURM 58799	HousingQA state-filtered retrieval at k=5 and k=10 after fixing the state-metadata casing bug.
SLURM 58871	CaseHOLD repaired top-k follow-up: `rag_simple` at k=1 and k=10 plus `rag_hyde` at k=5.
SLURM 58885	LegalBench-SCALR snap ablation: `rag_hyde` and `rag_snap_hyde_1call` at k=5, complementing existing `rag_simple` and `rag_snap_hyde_2call` rows.

BarExamQA already has full-corpus snap-HyDE evidence, so the immediate priority is not another broad BarExamQA sweep. The useful BarExamQA add-on is a narrower generated-query ablation, such as `multi_hyde_diverse`, if API budget remains after HousingQA and CaseHOLD complete.

6 Discussion: From Method Search to Bottleneck Routing

The most useful interpretation of these results is not that snap-HyDE is the final method. It is that a small family of fixed interventions can expose which part of a legal RAG pipeline is currently binding. This matters because many RAG papers collapse several different phenomena into one comparison: retrieval versus no retrieval. In our logs, that comparison is too coarse. LegalBench-SCALR loses heavily at top-1 and recovers at top-5, so a route that expands the candidate set is plausible. BarExamQA is top-k flat but improves with snap-conditioned final synthesis, so the route should preserve an answer frame rather than simply retrieve more passages. HousingQA likely requires metadata-constrained statutory retrieval, because the question state is part of the legal problem. CaseHOLD improves gold retrieval under two-call snap-HyDE but not answer accuracy, so the route should focus on mapping retrieved holdings back to answer options.

This gives a feasible version of the "agentic" idea without making the agent itself the unit of evaluation. Instead of asking a planner to invent an open-ended research workflow, we can ask a controller to choose among a small set of evidence-budgeted interventions. The controller would observe cheap features before or during retrieval: question type, answer format, available metadata, top-score margin, retrieved-document agreement, candidate-option similarity, and whether the snap answer agrees with retrieval. It would then choose a route: ordinary RAG, top-k expansion, snap-conditioned HyDE, state-filtered retrieval, multi-query diversity, or option-aware reranking. The result is still agentic in the sense that method adaptation happens on the fly, but it is experimentally cleaner because every route corresponds to a measured ablation cell.

Observed signal	Likely bottleneck	Route to test next
Top-1 fails, top-5 recovers, top-10 flat	Candidate-set size rather than answer reasoning	Use top-5 as default; spend budget on reranking or option grounding instead of deeper context.
Snap-only beats or matches retrieval	Evidence is distracting or under-specified relative to legal priors	Keep a first-pass legal frame in final synthesis and audit whether evidence overwrites correct priors.
Generated retrieval improves gold-hit but not answer accuracy	Evidence-to-answer conversion bottleneck	Add answer-option-aware synthesis, contradiction checks, or verifier scoring rather than more retrieval.
Question contains jurisdiction metadata	Corpus filtering and metadata normalization bottleneck	Filter before retrieval, validate non-empty evidence, and compare against generic top-k retrieval.
Retriever scores are low or retrieved passages disagree	Query formulation bottleneck	Try snap-HyDE or diverse HyDE passages, then measure whether gold retrieval and answer accuracy move together.

This framing also explains why the current negative results are useful. A negative CaseHOLD answer delta after a positive gold-hit delta is not just a failed method row; it identifies a downstream conversion problem. A flat BarExamQA top-k sweep does not make retrieval irrelevant; it says that retrieval depth is not the active lever for that slice. A pending HousingQA state-filter result is not a missing number to hide; it is exactly the kind of metadata gate a legal RAG system must satisfy before its answers can be trusted. These are the hooks for a stronger final project: tune snap-HyDE quality where query formulation is binding, tune metadata filters where jurisdiction is binding, and tune verifier or option-mapping logic where evidence conversion is binding.

The class-report claim is therefore deliberately modest but concrete. We built and audited a legal RAG harness that can run comparable methods across several legal benchmarks; we found one full-corpus legal snap-HyDE lift on BarExamQA; we found several $N = 200$ diagnostic signals that disagree with a single-method story; and we turned those signals into a routing agenda. That is a better foundation for future work than another broad sweep over loosely comparable prompts, because it says what should change when a method fails.

7 Limitations and Future Work

The report is intentionally conservative. Only BarExamQA has a full-corpus legal result. HousingQA, SCALR, and CaseHOLD are paired $N = 200$ diagnostic slices. These are useful for mechanism discovery and class-report evidence, but they are not final benchmark claims. HousingQA also has an unresolved metadata-filter gate: the first state-filter job produced empty retrieval and must not be cited as a method result.

The current method set is incomplete relative to the literature. We define Self-RAG, CRAG, and Speculative RAG, but do not reimplement them. We also do not yet log true Speculative-RAG draft/verifier metrics. The local HyDE methods need improvement: better corpus-style passage prompts, explicit jurisdiction handling for HousingQA, and answer-option-aware final synthesis for holding tasks. The goal is not to defend the current numbers as final; it is to use them to identify where tuning should be applied.

The next experiments should stay legal-only. First, finish HousingQA state-filtered retrieval and compare it to top-10 generic retrieval. Second, add SCALR `rag_hyde`, `rag_snap_hyde_1call`, and `multi_hyde_diverse` rows if API budget allows. Third, run repaired CaseHOLD top-1/top-5/top-10 after the Chroma repair so the depth-policy claim uses meaningful gold retrieval. Fourth, if we want a stronger retrieval benchmark, wire LegalBench-RAG or Legal RAG Bench rather than adding another generic multi-hop dataset.

8 Conclusion

The legal-only evidence supports a bottleneck-typed view of RAG. Snap-HyDE helps BarExamQA at full corpus, but BarExamQA is not retrieval-depth limited. SCALR is candidate-depth limited, HousingQA is a pending statutory metadata test, and CaseHOLD shows that better retrieval does not automatically convert to better answers. These mixed results are not a failure of the project. They are the project signal: before building a bigger agent or declaring a RAG method better, we need to know whether the legal task needs more candidates, better generated queries, state-aware filtering, answer anchoring, or a stronger evidence-to-option mapper.

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